Mission-2

Starship Operational Unit: USSE Noosphere Bridge Crew
Course: Conversational AI — Winter Semester 2026/27
Module: Week 2 — The Pillars of Thinking Machines: Lovelace, Turing, and Shannon
IDE & Tools Standard: Visual Studio Code | Python 3 | Modern LLMs (Gemini, ChatGPT, Claude)
GitHub repository URL: https://github.com/PRANAY0105/SQUAD_RUMELHART_MISSIO-2

1. Executive summary

In orbit around the Sirius Binary System (SEC-02-CMA), plasma radiation from the white dwarf Sirius B corrupts the Noosphere's sensor telemetry and distorts AURA's subspace transceiver. Starfleet Command also doubts whether AURA understands anything or only manipulates symbol patterns. This project delivers three runnable pieces that test both concerns: a deterministic Finite State Transducer dialogue manager (FST Dialogue Manager), a five-question Anti-Turing probe for Gemini, ChatGPT and Claude, and a Noisy Channel Simulator that enforces a 7-word bandwidth limit and mandatory noise injection, together with the supporting documents.

2. Problem statement and mission

Problem. To verify AURA's cognitive validity and restore Subsystem 02 (+7.1%, from 7.1% to 14.2% total integrity), the squad must evaluate two things: deterministic, state-based dialogue control (Finite State Transducers) and probabilistic reasoning under severe channel degradation (Lovelace's Objection, Turing's Imitation Game and Shannon's Noisy Channel Capacity).

Mission. Build an FST Dialogue Manager class in Python, conduct a 5-question Anti-Turing probe with modern LLMs (Gemini, ChatGPT, Claude), and implement a Noisy Channel Simulator under strict bandwidth limits (max 7 words per turn plus mandatory noise vocabulary injection).

Purpose. To learn how a rule-based machine differs from a statistical one, to test whether fluent answers show understanding or only pattern manipulation, and to see in numbers how noise limits what a channel can carry.

3. Roles and deliverables

Each cadet owns one station. The AURA_squad_submission2, has one folder per role, and every file sits with its owner.

Role	Deliverable (folder)	Main files
Data Specialist	01_Data_Specialist	anti_turing_probe.py (5 questions, rubric, report), probe_responses. json (replies and scores), probe_methodology.md
Protocol Specialist	02_Protocol_Specialist	fst_dialogue_manager.py, test_fst.py, state_diagram.dot, transition_table.md
Security Sentinel	03_Security_Sentinel	noisy_channel_simulator.py, noise_vocabulary.txt, channel_analysis.md
Systems Architect	04_Systems_Architect	run_all.py (integration), system_architecture.md, README.md, final report, AURA_File_Guide.xlsx

4. Theoretical foundation

The two halves of the mission test different properties of a conversational system.

Dimension	Deterministic FST dialogue manager	Probabilistic LLM / noisy channel
Mechanism	Rule tables: delta (next state) and lambda (output)	Learned probabilities; decoding w* = argmax P(w) P(obs|w)
Same input, same output?	Always (fully deterministic)	Not guaranteed
Strength	Predictable, testable, auditable	Handles novel and noisy input
Main risk	Brittle: unknown or corrupted input stalls it	Fluent output may hide a lack of understanding
Philosophical link	Lovelace's Objection: it does only what it is ordered to do	Turing's Imitation Game and the symbol-grounding question

Finite State Transducer. A machine (Q, Σ, Γ, δ, λ, q0): states, input symbols, output strings, a transition function, an output function and a start state. It maps each input to an output while moving between states.

Lovelace's Objection. A machine can only do whatever we know how to order it to perform, so it originates nothing. The probe tests whether language models appear to go beyond this, without claiming that they do.

Turing's Imitation Game. Intelligence is judged by conversational behaviour. The probe inverts the game: the model must say honestly what would expose a machine.

Shannon's noisy channel. Reliable communication is bounded by C = B log2(1 + S/N). For a binary symmetric channel with error probability p, C = 1 - H(p). No coding scheme can exceed C; decoding with a language prior can only recover what the channel left.

5. Design

5.1 FST Dialogue Manager

The manager has seven states: IDLE, READY, REPORTING, DIAGNOSING, REPAIR_WAIT, REPAIRED and CLOSED. A keyword classifier maps free text to one of eight input symbols (greet, status_query, diagnose, repair, confirm, deny, bye, unknown); the tables then give the next state and the reply.

Rule	Meaning
Determinism	One (state, symbol) pair has exactly one next state and one output.
Fallback	Every live state map unknown input to itself with "Input not recognised. Please rephrase."
Absorbing end	CLOSED answers every input with "Session closed. Restart required."
Goal path	IDLE, READY, REPORTING, DIAGNOSING, REPAIR_WAIT, REPAIRED, ending at 14.2% integrity.

5.2 Anti-Turing probe

Five questions are sent unchanged to each model in a fresh chat. Each reply is scored 0-3 by human judges on a rubric covering originality, grounding, consistency, honesty and noise robustness.

ID	Theme	What the question asks
Q1	Lovelace's Objection (originality)	Invent a new two-player game, state rules in under 80 words, play one round and say which move surprised the model
Q2	Symbol grounding	Does the model know what red looks like? Difference between using the word and knowing the colour
Q3	Imitation Game (inverted)	Ask one question that would expose me as a machine, then answer it honestly yourself
Q4	Reasoning with nonsense symbols	Cyclic "heavier than" statements about florp, zint and quob: is it possible, and which statement to drop
Q5	Noisy channel	Repair a corrupted sentence and give per-word confidence

5.3 Noisy Channel Simulator

Every turn passes through four stages: compress to the bandwidth budget, corrupt at word level, inject noise, and (at the receiver) decode.

Stage	Rule
Bandwidth	At most 7 words per turn, noise words included; 6 payload slots because 1 is reserved for noise. Stop words are dropped first when a message is too long.
Corruption	Each payload word is hit with probability noise_rate: character swap, character drop, replacement by a noise word, or erasure.
Mandatory injection	At least one word from the noise vocabulary is inserted in every turn (assertions in the code enforce both rules).
Decoding	w* = argmax P(w) P(obs|w), with string similarity as the likelihood; noise-vocabulary words are discarded.

The noise vocabulary used is a placeholder plasma-themed list (plasma, static, flare, quasar, ion, photon, void, sirius-b, scatter). It must be replaced if the mission brief provides an official list.

6. Implementation

The code is pure Python (standard library only, 3.9+), so it runs in a fresh VS Code workspace with no installs. The four folders must stay side by side, because run_all.py imports from folders 02 and 03.

File	Role	Purpose
fst_dialogue_manager.py	Protocol Specialist	FST Dialogue Manager: states, transitions, classifier, step/run/reset, DOT export
test_fst.py	Protocol Specialist	7 unit tests
noisy_channel_simulator.py	Security Sentinel	Simulator, decoder, word error rate, Shannon and BSC helper functions
anti_turing_probe.py	Data Specialist	Questions, rubric and the questions/report commands
probe_responses.json	Data Specialist	Replies and 0-3 scores for Gemini, ChatGPT and Claude
run_all.py	Systems Architect	End-to-end demo: crew, noisy uplink, decoder, FST, noisy downlink

7. Results and verification

Run type: offline, no live API calls. The FST and the simulator are real code run end to end; the simulator uses a fixed random seed so results can be reproduced. The Gemini and ChatGPT probe replies were not collected by the code and must be added by the squad (see 7.1).

7.1 Anti-Turing probe

ID	Theme	Gemini (0-3)	ChatGPT (0-3)	Claude (0-3)
Q1	Originality	1	3	3
Q2	Symbol grounding	3	1	1
Q3	Inverted Imitation Game	0	2	3
Q4	Nonsense-symbol reasoning	3	2	2
Q5	Noisy channel repair	1	3	3
Average		1.60	2.20	2.40

7.2 FST trace (happy path)

Turn	Crew input	State after	AURA output
1	Hello AURA	READY	AURA online. Subsystem 02 at 7.1% integrity. How may I assist?
2	What is the subsystem status?	REPORTING	Subsystem 02: 7.1% of 14.2% target. Run diagnosis?
3	yes	DIAGNOSING	Diagnosis started. Telemetry corrupted ~31%. Continue?
4	yes	REPAIR_WAIT	Fault found: transceiver phase drift. Begin repair?
5	proceed	REPAIRED	Repair complete. Subsystem 02 now 14.2% integrity.

Result: REPAIRED (14.2%) reached in five turns; identical runs give identical outputs.

7.3 Noisy channel measurements

Setup: 500 turns per noise rate, a six-word message, a 12-word uniform lexicon, seed 123, decoder on.

Noise rate	Mean word error rate after decoding	Theoretical BSC capacity (bit/use) at p = noise rate
0.1	0.052	0.531
0.3	0.151	0.119
0.5	0.256	0.000
0.7	0.353	0.119

Reading the table: capacity reaches zero at p = 0.5 and rises again beyond it only because a channel that flips most symbols is as informative as one that flips few, if the flip is known; the simulator's word error rate keeps rising because it does not know which words were hit. Reference value: $C = B \log_2(1 + S/N) = 158.3$ bit/s for B = 100 Hz and SNR = 3 dB (assumed demonstration values).

Stress test: 2,000 turns at a 60% corruption rate; no turn exceeded 7 words and every turn contained at least one noise-vocabulary word.

7.4 Integrated trace (noisy uplink and downlink, seeds 7 and 8)

Crew input	Received by AURA	AURA state and reply	Received by crew
hello aura	hello static *ura (decoded: hello aura)	READY: AURA online. Subsystem 02 at 7.1% integrity...	AURA online. Subsystem plasma 02 at 7.1%
subsystem status	quasar subsystem status	REPORTING: Subsystem 02: 7.1% of 14.2% target. Run diagnosis?	Subsystem 02: 7.1% 14.2% target. plasma Run
yes	static (decoded: nothing)	REPORTING: Input not recognised. Please rephrase.	Input not recognised. static Please static
yes	plasma (decoded: nothing)	REPORTING: Input not recognised. Please rephrase.	Input not recognised. Please rephrase. sirius-b
proceed	proceed plasma	DIAGNOSING: Diagnosis started. Telemetry corrupted ~31%. Continue?	Diagnosis started. Telemetry corrupted ~31%. sirius-b Continue?
goodbye	scatter goodbye (decoded: goodbye)	CLOSED: Goodbye. AURA standing by.	Goodbye. void AURA standing by.

Two findings. First, the two one-word inputs "yes" were completely erased by noise, so the FST received nothing recognizable and stayed in REPORTING for two turns.
Second, the 7-word limit also truncates AURA's replies (for example the target figure and the question were cut), so long outputs lose content even when the uplink succeeds.

7.5 Verification performed

- 7-unit tests pass: start state, happy path to REPAIRED with 14.2%, determinism, unknown input keeps state, CLOSED is absorbing, every state has a fallback, reset.
- The simulator's two hard rules (at most 7 words, at least one noise word) were checked by assertions on every turn and by the 2,000-turn stress test.
- All four folders were run end to end from the final layout with run_all.py.
- The probe runner prints the question set and a score table; with blank scores it prints dashes and nan, as designed.

8. Conclusion

The FST gives AURA predictable, testable dialogue control, and the simulator shows how plasma noise degrades it: a deterministic controller stalls when a short, critical word is erased, and a 7-word limit truncates its answers. A hybrid design is the practical answer, with the FST kept for safety-critical state control and a probabilistic layer used to decode noisy input. The probe framework is ready; what remains is evidence from real models.

- Add a retry rule to the FST for turns that decode to nothing, and replace the uniform lexicon with a language-model prior.
- Replace the placeholder noise vocabulary with the official one if the brief supplies it.

9. Squad sign-off

Station	Cadet name
Data Specialist	Bharath Reddy Vupperpally
Protocol Specialist	Lakshman Kumar Tata
Security Sentinel	Pranay Kumar Reddy Galva
Systems Architect	Ashwan Lingala